

UNIVERSITY OF WAH(UOW)
DEPARTMENT OF COMPUTER
SCIENCE



Lab Task # 02

COURSE:

Information & Security Lab

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Task # 01

```
Id = []
for i in range(10):
    j = input("Enter Student ID's: ")
    Id.append(j)

Uni = []
count = 0
for i in Id:
    if i == "0":
        break
    if i in Uni:
        print(i,"Warning of this studnet")
    else:
        Uni.append(i)
        count = count + 1

for j in Uni:
    print(j)
print("Total Number of Student ID's:",count)
```

Output

```
Enter Student ID's: 111
Enter Student ID's: 222
Enter Student ID's: 222
Enter Student ID's: 333
Enter Student ID's: 333
Enter Student ID's: 555
Enter Student ID's: 777
Enter Student ID's: 0
Enter Student ID's: 121
Enter Student ID's: 334
222 Warning of this studnet
333 Warning of this studnet
111
222
333
555
777
Total Number of Student ID's: 7
```

Task # 02

```

# ----- Task 2 -----
password = input("Enter Password: ")

while len(password) <= 8:
    print("Password must have 8 charachters.")
    password = input("Enter Password: ")

special_char = "!@#$%&*"

has_char = any(ch.isupper() for ch in password)
has_number = any(ch.isdigit() for ch in password)
has_special = any(ch in special_char for ch in password)

if has_char and has_number and has_special:
    print("Password is Valid.")
else:
    if not has_char:
        print("Password must have a character.")
    if not has_number:
        print("Password must have a number.")
    if not has_special:
        print("Password must have any special character.")

```

Output

```

Enter Password: Yxajx%^xl888
Password is Valid.
PS D:\University Data\3rd Semester\Information & Security Lab\Lab work>

```

Task # 03

```
# ----- Task 3 -----
students = []

def add_student(name, *marks):
    student = {"name": name, "marks": marks}
    students.append(student)

def calculate_average(*marks):
    return sum(marks) / len(marks)

def assign_grade(avg):
    if avg >= 90:
        return "A"
    elif avg >= 80:
        return "B"
    elif avg >= 70:
        return "C"
    elif avg >= 60:
        return "D"
    else:
        return "F"
```

```
for i in range(5):
    print(f"\nStudent {i + 1}")
    name = input("Enter name: ")
    marks = []
    for j in range(3):
        mark = float(input(f"Enter mark {j + 1}: "))
        marks.append(mark)
    add_student(name, *marks)

print("\nStudent Report")
print(f"{'Name':<15}{'Average':<10}{'Grade'}")
print("-" * 30)

for s in students:
    avg = calculate_average(*s["marks"])
    grade = assign_grade(avg)
    print(f"{s['name']:<15}{avg:<10.2f}{grade}")
```

Output

Enter mark 3: 55

Enter mark 3: 22

22	22.00	F
----	-------	---

Task # 4

```

def get_max_min(temps):
    max_temp = temps[0]
    min_temp = temps[0]
    for t in temps:
        if t > max_temp:
            max_temp = t
        if t < min_temp:
            min_temp = t
    return max_temp, min_temp

def above_average(temps):
    Click to collapse the range.
    for t in temps:
        total += t
    avg = total / len(temps)
    print(f"\nAverage temperature: {avg:.2f}")
    print("Days with above-average temperature:")
    for i in range(len(temps)):
        if temps[i] > avg:
            print(f"Day {i + 1}: {temps[i]}")

temps = []
print("Enter temperatures for 7 days:")
for i in range(7):
    t = int(input(f"Day {i + 1}: "))
    temps.append(t)

max_t, min_t = get_max_min(temps)
print(f"\nHighest temperature: {max_t}")
print(f"Lowest temperature: {min_t}")

above_average(temps)

```

Output

```

Enter temperatures for 7 days:
Day 1: 22
Day 2: 44
Day 3: 66
Day 4: 44
Day 5: 11
Day 6: -9
Day 7: -1

Highest temperature: 66
Lowest temperature: -9

Average temperature: 25.29
Days with above-average temperature:
Day 2: 44
Day 3: 66
Day 4: 44

```

Task # 05

```
class Book:
    def __init__(self, title, author):
        self.title = title
        self.author = author
        self.is_borrowed = False

    def borrow_book(self):
        if self.is_borrowed == False:
            self.is_borrowed = True
            print("You borrowed", self.title)
        else:
            print("Already borrowed")

    def return_book(self):
        if self.is_borrowed == True:
            self.is_borrowed = False
            print("You returned", self.title)
        else:
            print("It was not borrowed")

    def display_info(self):
        if self.is_borrowed == True:
            status = "Borrowed"
        else:
            status = "Available"
        print(self.title, "-", self.author, "-", status)
```

```

books = [
    Book("English", "Ali"),
    Book("Islamiyat", "Atif"),
    Book("Science", "Rafay"),
    Book("Physics", "James Bond"),
    Book("Mathematics", "Shilaa ki jawani")
]
while True:
    print("\n1. Show books")
    print("2. Borrow")
    print("3. Return")
    print("4. Exit")
    ch = input("Enter choice: ")
    if ch == "1":
        for i in range(len(books)):
            print(i+1, end=". ")
            books[i].display_info()
    elif ch == "2":
        n = int(input("Enter book number: ")) - 1
        books[n].borrow_book()
    elif ch == "3":
        n = int(input("Enter book number: ")) - 1
        books[n].return_book()
    elif ch == "4":
        break
    else:
        print("Wrong choice")

```

Output

```

1. Show books
2. Borrow
3. Return
4. Exit
Enter choice: 1
1. English - Ali - Available
2. Islamiyat - Atif - Available
3. Science - Rafay - Available
4. Physics - James Bond - Available
5. Mathematics - Shilaa ki jawani - Available

```

Task # 06

```
# -----6-----
class Employee:
    def __init__(self, name, dept):
        self.name = name
        self.dept = dept
        self.monthly_scores = []

    def add_score(self, score):
        self.monthly_scores.append(score)

    def average_score(self):
        total = 0
        for s in self.monthly_scores:
            total += s
        avg = total / len(self.monthly_scores)
        return avg

    def performance_level(self):
        avg = self.average_score()
        if avg >= 90:
            return "Excellent"
        elif avg >= 75:
            return "Good"
        elif avg >= 60:
            return "Average"
        else:
            return "Poor"
```

```
emp1 = Employee("Alice", "HR")
emp2 = Employee("Bob", "Sales")
emp3 = Employee("Charlie", "IT")
emps = [emp1, emp2, emp3]
import random
for e in emps:
    for i in range(5): # 5 months
        e.add_score(random.randint(50, 100))
for e in emps:
    print("\nName:", e.name)
    print("Dept:", e.dept)
    print("Average Score:", round(e.average_score(), 2))
    print("Performance:", e.performance_level())
```

Output

Name: Alice
Dept: HR
Average Score: 71.8
Performance: Average

Name: Bob
Dept: Sales
Average Score: 84.0
Performance: Good

Name: Charlie
Dept: IT
Average Score: 72.0
Performance: Average

PS D:\University Data\3rd Semester\Information & Security Lab\Lab work> █